D3 Sigma notation A level only

Name:

Class:

Date:

Time: **80 minutes**

Marks: **67 marks**

Comments:

Q1.

(a) A geometric series begins 420 + 294 + 205.8 + ….

(i) Show that the common ratio of the series is 0.7.

(1)

(ii) Find the sum to infinity of the series.

(2)

(iii) Write the nth term of the series in the form $p \times q^n$, where p and q are constants.

(2)

(b) The first term of an arithmetic series is 240 and the common difference of the series is −8.

The nth term of the series is u_n.

(i) Write down an expression for u_n.

(1)

(ii) Given that $u_k = 0$, find the value of $\sum_{n=1}^{k} u_n$.

(4)

(Total 10 marks)

Q2.

An arithmetic series has first term a and common difference d.

The sum of the first 25 terms of the series is 3500.

(a) Show that $a + 12d = 140$.

(3)

(b) The fifth term of this series is 100.

Find the value of d and the value of a.

(4)

(c) The nth term of this series is u_n. Given that

$$33\left(\sum_{n=1}^{25} u_n - \sum_{n=1}^{k} u_n\right) = 67\sum_{n=1}^{k} u_n$$

find the value of $\sum_{n=1}^{k} u_n$.

(3)
(Total 10 marks)

Q3.

An arithmetic series has first term a and common difference d.

The sum of the first 31 terms of the series is 310.

(a) Show that $a + 15d = 10$.

(3)

(b) Given also that the 21st term is twice the 16th term, find the value of d.

(3)

(c) The nth term of the series is u_n. Given that $\sum_{n=1}^{k} u_n = 0$, find the value of k.

(4)
(Total 10 marks)

Q4.

(a) An infinite geometric series has common ratio r.

The first term of the series is 10 and its sum to infinity is 50.

(i) Show that $r = \frac{4}{5}$.

(2)

(ii) Find the second term of the series.

(2)

(b) The first and second terms of the geometric series in part (a) have the same values as the 4th and 8th terms respectively of an arithmetic series.

(i) Find the common difference of the arithmetic series.

(3)

(ii) The nth term of the arithmetic series is u_n. Find the value of $\sum_{n=1}^{40} u_n$.

(4)
(Total 11 marks)

Q5.

A geometric series has second term 375 and fifth term 81.

(a) (i) Show that the common ratio of the series is 0.6.

(3)

(ii) Find the first term of the series.

(2)

(b) Find the sum to infinity of the series.

(2)

(c) The nth term of the series is u_n. Find the value of $\sum_{n=6}^{\infty} u_n$.

(4)
(Total 11 marks)

Q6.

The nth term of a geometric sequence is u_n, where

$$u_n = 3 \times 4^n$$

(a) Find the value of u_1 and show that $u_2 = 48$.

(2)

(b) Write down the common ratio of the geometric sequence.

(1)

(c) (i) Show that the sum of the first 12 terms of the geometric sequence is $4^k - 4$, where k is an integer.

(3)

(ii) Hence find the value of $\sum_{n=2}^{12} u_n$.

(1)
(Total 7 marks)

Q7.

An arithmetic series has fifth term 46 and twentieth term 181.

(a) (i) Show that the common difference is 9.

(3)

(ii) Find the first term.

(1)

(b) Find the sum of the first 20 terms of the series.

(2)

(c) The nth term of the series is u_n .

Given that the sum of the first 50 terms of the series is 11525, find the value of

$$\sum_{n=21}^{50} u_n$$

(2)
(Total 8 marks)